

Shah Md. Tasrif Rahman

* Encoding: UTF-8.

```
PRESERVE.  
SET DECIMAL DOT.
```

* Since I messed up while generating the output of my first assignment, I'll be properly segmenting my code so that the final output is clean and reproducible using whatever syntax I used during data analysis.

* /Step-1: Data loading/

```
GET DATA /TYPE=TXT  
/FILE="E:\Statistical Softwares\SPSS\CUSPC-SPSS\Class-7\salary regression data.csv"  
/ENCODING='UTF8'  
/DELIMITERS=","  
/QUALIFIER='''  
/ARRANGEMENT=DELIMITED  
/FIRSTCASE=2  
/DATATYPEMIN PERCENTAGE=95.0  
/VARIABLES=  
employee_id AUTO  
age AUTO  
education_years AUTO  
experience_years AUTO  
training_hours AUTO  
region AUTO  
annual_salary_k AUTO  
/MAP.  
RESTORE.  
CACHE.  
EXECUTE.
```

Data written to the working file.
7 variables and 250 cases written.

Variable: employee_id	Type: Number	Format : F4
Variable: age	Type: Number	Format : F2
Variable: education_years	Type: Number	Format : F4.1
Variable: experience_years	Type: Number	Format : F4.1
Variable: training_hours	Type: Number	Format : F2
Variable: region	Type: String	Format : A8
Variable: annual_salary_k	Type: Number	Format : F4.1

Substitute the following to build syntax for these data.

```
/VARIABLES=  
employee_id F4  
age F2  
education_years F4.1  
experience_years F4.1  
training_hours F2  
region A8  
annual_salary_k F4.1
```

```
DATASET NAME DataSet1 WINDOW=FRONT.
```

* /Step-2: Data labelling/

```
VARIABLE LABELS  
employee_id 'Anonymised employee identifier'  
age 'Age at time of extract (Years)'  
education_years 'Total years of completed formal education (Years)'  
experience_years 'Total years of professional work experience (Years)'  
training_hours 'Company-funded training received in the last 12 months (Hours)'  
region 'Office location type (Urban/Suburban/Rural)'
```

```

        annual_salary_k 'Gross annual salary (USD thousands)'.
EXECUTE.

VARIABLE LEVEL employee_id (NOMINAL).
EXECUTE.

* /Step-3: Data Checking/

DATASET ACTIVATE DataSet1.
FREQUENCIES VARIABLES=employee_id age education_years experience_years training_hours
region
        annual_salary_k
/FORMAT=NOTABLE
/STATISTICS=RANGE MINIMUM MAXIMUM
/ORDER=ANALYSIS.

```

Frequencies

[DataSet1]

		Statistics						
		Anonymised employee identifier	Age at time of extract (Years)	Total years of completed formal education (Years)	Total years of professional work experience (Years)	Company-funded training received in the last 12 months (Hours)	Office location type (Urban/Suburban/ Rural)	Gross annual salary (USD thousands)
N	Valid	250	250	250	250	250	250	250
	Missing	0	0	0	0	0	0	0
Range		249	24	11.6	19.6	75		50.2
Minimum		5001	22	10.1	.0	0		37.2
Maximum		5250	46	21.7	19.6	75		87.4

```

FREQUENCIES VARIABLES=region
/ORDER=ANALYSIS.

```

Frequencies

Statistics

Office location type

(Urban/Suburban/Rural)

N	Valid	250
	Missing	0

Office location type (Urban/Suburban/Rural)

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Rural	44	17.6	17.6	17.6
	Suburban	75	30.0	30.0	47.6
	Urban	131	52.4	52.4	100.0
	Total	250	100.0	100.0	

*** All 250 observations are present in the given dataset, and no impossible values were found. We can safely proceed to the next steps of analysis.**

* /Task 1/

.

* Custom Tables.

CTABLES

/VLABELS VARIABLES=age education_years experience_years training_hours

annual_salary_k

DISPLAY=LABEL

/TABLE age [COUNT 'Complete cases (n)' F40.0, MINIMUM, MAXIMUM, MEAN, MEDIAN, STDDEV]

+

education_years [COUNT 'Complete cases (n)' F40.0, MINIMUM, MAXIMUM, MEAN, MEDIAN, STDDEV] +

experience_years [COUNT 'Complete cases (n)' F40.0, MINIMUM, MAXIMUM, MEAN, MEDIAN, STDDEV] +

training_hours [COUNT 'Complete cases (n)' F40.0, MINIMUM, MAXIMUM, MEAN, MEDIAN, STDDEV] +

annual_salary_k [COUNT 'Complete cases (n)' F40.0, MINIMUM, MAXIMUM, MEAN, MEDIAN, STDDEV]

/CRITERIA CILEVEL=95.

Custom Tables

	Complete cases (n)	Minimum	Maximum	Mean	Median	Standard Deviation
Age at time of extract (Years)	250	22	46	30	30	6
Total years of completed formal education (Years)	250	10.1	21.7	16.1	16.3	2.3
Total years of professional work experience (Years)	250	.0	19.6	7.7	7.7	4.9
Company-funded training received in the last 12 months (Hours)	250	0	75	31	31	15
Gross annual salary (USD thousands)	250	37.2	87.4	61.8	61.7	10.1

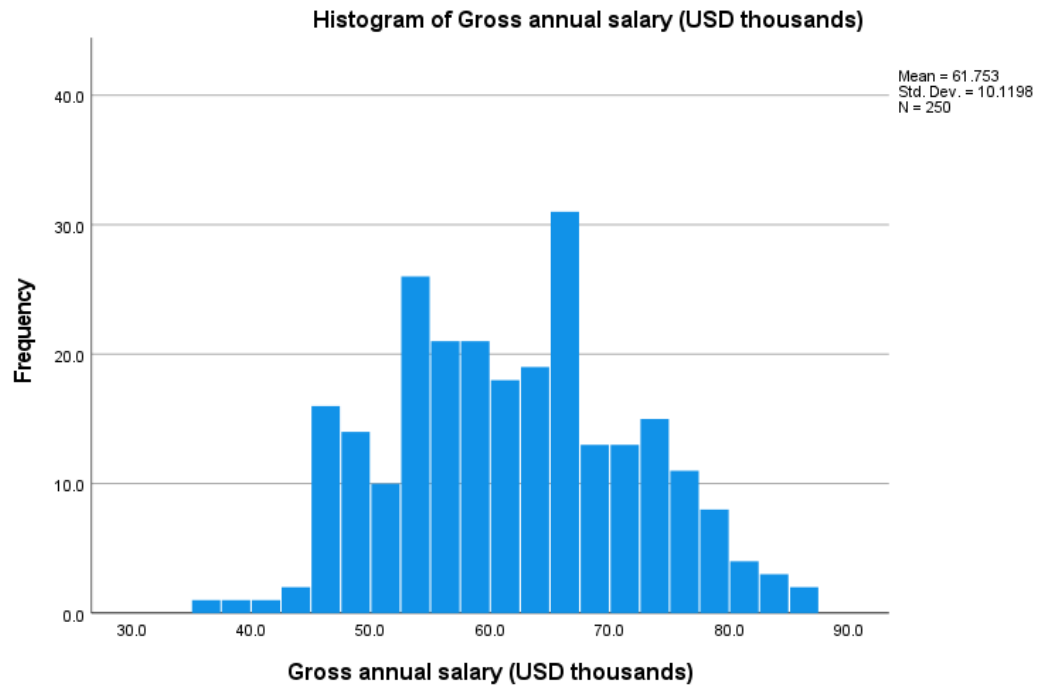
* /Task 2/

* /Graph-2a/

* Chart Builder.

```
GGRAPH
  /GRAPHDATASET NAME="graphdataset" VARIABLES=annual_salary_k MISSING=LISTWISE
REPORTMISSING=NO
  /GRAPHSPEC SOURCE=INLINE
  /FRAME OUTER=NO INNER=NO
  /GRIDLINES XAXIS=NO YAXIS=YES
  /STYLE GRADIENT=NO.
BEGIN GPL
  SOURCE: s=userSource(id("graphdataset"))
  DATA: annual_salary_k=col(source(s), name("annual_salary_k"))
  GUIDE: axis(dim(1), label("Gross annual salary (USD thousands)"))
  GUIDE: axis(dim(2), label("Frequency"))
  GUIDE: text.title(label("Histogram of Gross annual salary (USD thousands)"))
  ELEMENT: interval(position(summary.count(bin.rect(annual_salary_k))),
    shape.interior(shape.square))
END GPL.
```

GGraph



* /Graph-2b/

* Chart Builder.

GGRAPH

/GRAPHDATASET NAME="graphdataset" VARIABLES=region COUNT() [name="COUNT"]

MISSING=LISTWISE

REPORTMISSING=NO

/GRAPHSPEC SOURCE=INLINE

/FRAME OUTER=NO INNER=NO

/GRIDLINES XAXIS=NO YAXIS=YES

/STYLE GRADIENT=NO.

BEGIN GPL

SOURCE: s=userSource(id("graphdataset"))

DATA: region=col(source(s), name("region"), unit.category())

DATA: COUNT=col(source(s), name("COUNT"))

GUIDE: axis(dim(1), label("Office location type (Urban/Suburban/Rural)"))

GUIDE: axis(dim(2), label("Percent"))

GUIDE: text.title(label("Bar chart of Office location type (Urban/Suburban/Rural)"))

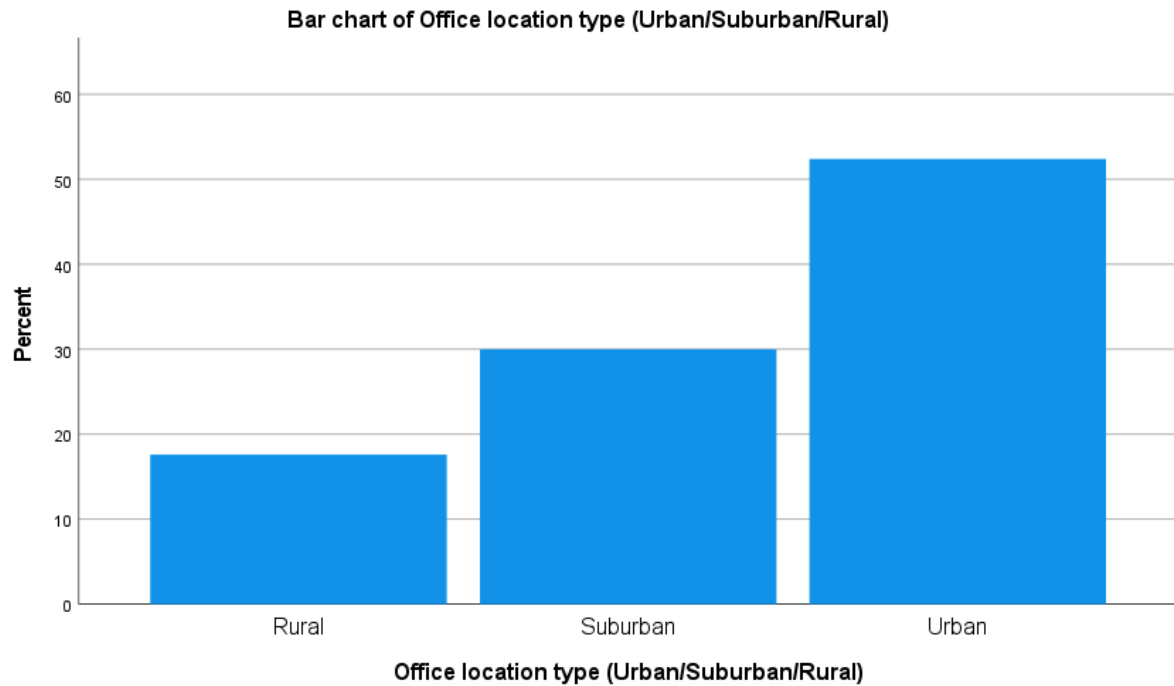
SCALE: linear(dim(2), include(0))

ELEMENT: interval(position(summary.percent(region*COUNT,
base.all(acrossPanels()))),

shape.interior(shape.square))

END GPL.

GGraph



```
* /Graph-2c/
```

```
* Chart Builder.
```

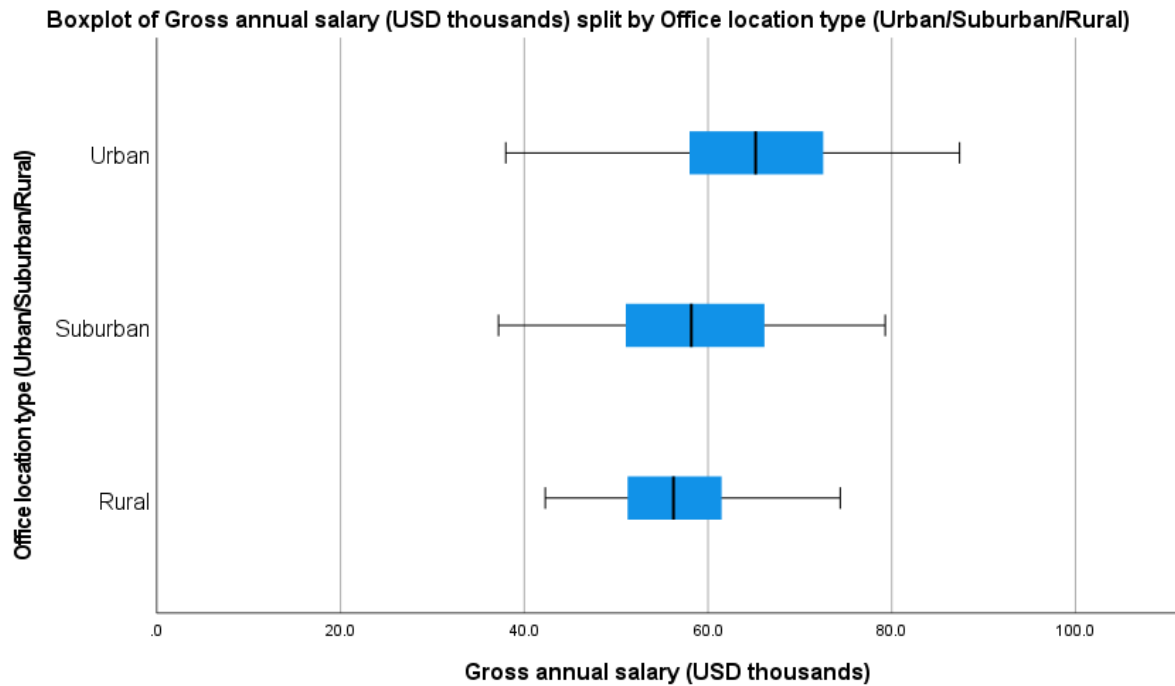
```
GGRAPH
```

```
  /GRAPHDATASET NAME="graphdataset" VARIABLES=region annual_salary_k MISSING=LISTWISE  
    REPORTMISSING=NO  
  /GRAPHSPEC SOURCE=INLINE  
  /FRAME OUTER=NO INNER=NO  
  /GRIDLINES XAXIS=NO YAXIS=YES  
  /STYLE GRADIENT=NO.
```

```
BEGIN GPL
```

```
  SOURCE: s=userSource(id("graphdataset"))  
  DATA: region=col(source(s), name("region"), unit.category())  
  DATA: annual_salary_k=col(source(s), name("annual_salary_k"))  
  DATA: id=col(source(s), name("$CASENUM"), unit.category())  
  COORD: transpose()  
  GUIDE: axis(dim(1), label("Office location type (Urban/Suburban/Rural)"))  
  GUIDE: axis(dim(2), label("Gross annual salary (USD thousands)"))  
  GUIDE: text.title(label("Boxplot of Gross annual salary (USD thousands) split by  
Office location type (Urban/Suburban/Rural)"))  
  SCALE: linear(dim(2), include(0))  
  ELEMENT: schema(position(bin.quantile.letter(region*annual_salary_k)), label(id))  
END GPL.
```

GGraph

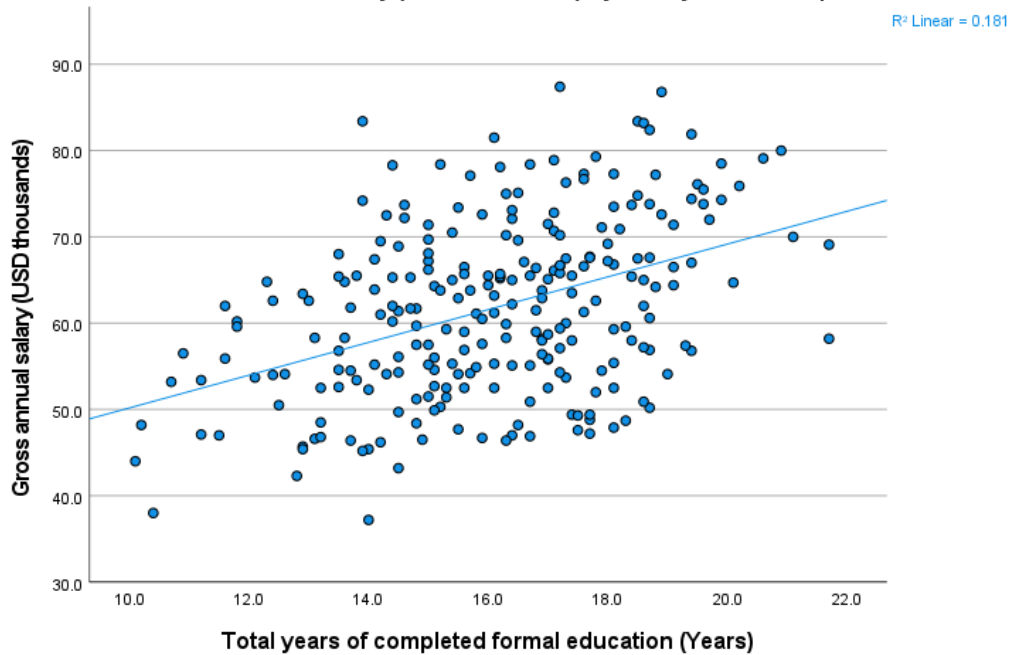


```
* /Graph-2d_1/

* Chart Builder.
GGRAPH
  /GRAPHDATASET NAME="graphdataset" VARIABLES=education_years annual_salary_k
MISSING=LISTWISE
  REPORTMISSING=NO
  /GRAPHSPEC SOURCE=INLINE
  /FITLINE TOTAL=YES SUBGROUP=NO
  /FRAME OUTER=NO INNER=NO
  /GRIDLINES XAXIS=NO YAXIS=YES
  /STYLE GRADIENT=NO.
BEGIN GPL
  SOURCE: s=userSource(id("graphdataset"))
  DATA: education_years=col(source(s), name("education_years"))
  DATA: annual_salary_k=col(source(s), name("annual_salary_k"))
  GUIDE: axis(dim(1), label("Total years of completed formal education (Years)"))
  GUIDE: axis(dim(2), label("Gross annual salary (USD thousands)"))
  GUIDE: text.title(label("Scatter Plot of Gross annual salary (USD thousands) by Total
years of ",
  "completed formal education (Years)"))
  ELEMENT: point(position(education_years*annual_salary_k))
END GPL.
```

GGraph

Scatter Plot of Gross annual salary (USD thousands) by Total years of completed formal education (Years)



```
* /Graph-2d_2/
```

```
* Chart Builder.
```

```
GGRAPH
```

```
  /GRAPHDATASET NAME="graphdataset" VARIABLES=experience_years annual_salary_k
```

```
MISSING=LISTWISE
```

```
  REPORTMISSING=NO
```

```
  /GRAPHSPEC SOURCE=INLINE
```

```
  /FITLINE TOTAL=YES SUBGROUP=NO
```

```
  /FRAME OUTER=NO INNER=NO
```

```
  /GRIDLINES XAXIS=NO YAXIS=YES
```

```
  /STYLE GRADIENT=NO.
```

```
BEGIN GPL
```

```
  SOURCE: s=userSource(id("graphdataset"))
```

```
  DATA: experience_years=col(source(s), name("experience_years"))
```

```
  DATA: annual_salary_k=col(source(s), name("annual_salary_k"))
```

```
  GUIDE: axis(dim(1), label("Total years of professional work experience (Years)"))
```

```
  GUIDE: axis(dim(2), label("Gross annual salary (USD thousands)"))
```

```
  GUIDE: text.title(label("Scatter Plot of Gross annual salary (USD thousands) by Total years of ",
```

```
    "professional work experience (Years)"))
```

```
  ELEMENT: point(position(experience_years*annual_salary_k))
```

```
END GPL.
```


GGraph

Scatter Plot of Gross annual salary (USD thousands) by Total years of professional work experience (Years)



```
* /Graph-2d_3/
```

```
* Chart Builder.
```

```
GGRAPH
```

```
  /GRAPHDATASET NAME="graphdataset" VARIABLES=training_hours annual_salary_k
```

```
MISSING=LISTWISE
```

```
  REPORTMISSING=NO
```

```
  /GRAPHSPEC SOURCE=INLINE
```

```
  /FITLINE TOTAL=YES SUBGROUP=NO
```

```
  /FRAME OUTER=NO INNER=NO
```

```
  /GRIDLINES XAXIS=NO YAXIS=YES
```

```
  /STYLE GRADIENT=NO.
```

```
BEGIN GPL
```

```
  SOURCE: s=userSource(id("graphdataset"))
```

```
  DATA: training_hours=col(source(s), name("training_hours"))
```

```
  DATA: annual_salary_k=col(source(s), name("annual_salary_k"))
```

```
  GUIDE: axis(dim(1), label("Company-funded training received in the last 12 months  
(Hours)"))
```

```
  GUIDE: axis(dim(2), label("Gross annual salary (USD thousands)"))
```

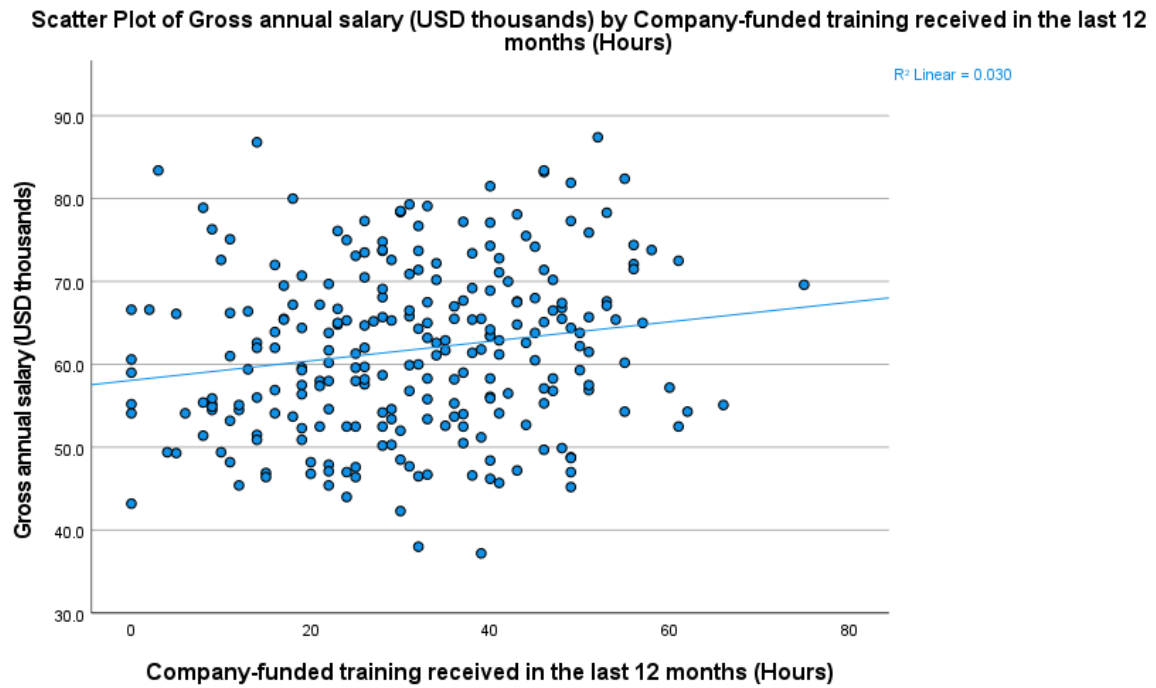
```
  GUIDE: text.title(label("Scatter Plot of Gross annual salary (USD thousands) by  
Company-funded ",
```

```
    "training received in the last 12 months (Hours)"))
```

```
  ELEMENT: point(position(training_hours*annual_salary_k))
```

```
END GPL.
```

GGraph



* /Task 3/

* /Main Table/

CORRELATIONS

/VARIABLES=age education_years experience_years training_hours annual_salary_k

/PRINT=TWOTAIL NOSIG FULL

/MISSING=PAIRWISE.

Correlations

Correlations						
		Age at time of extract (Years)	Total years of completed formal education (Years)	Total years of professional work experience (Years)	Company-funded training received in the last 12 months (Hours)	Gross annual salary (USD thousands)
Age at time of extract (Years)	Pearson Correlation	1	.420**	.785**	.098	.624**
	Sig. (2-tailed)		.000	.000	.123	.000
	N	250	250	250	250	250
Total years of completed formal education (Years)	Pearson Correlation	.420**	1	.007	-.031	.425**
	Sig. (2-tailed)	.000		.910	.630	.000
	N	250	250	250	250	250
Total years of professional work experience (Years)	Pearson Correlation	.785**	.007	1	.126*	.609**
	Sig. (2-tailed)	.000	.910		.046	.000
	N	250	250	250	250	250
Company-funded training received in the last 12 months (Hours)	Pearson Correlation	.098	-.031	.126*	1	.173**
	Sig. (2-tailed)	.123	.630	.046		.006
	N	250	250	250	250	250
Gross annual salary (USD thousands)	Pearson Correlation	.624**	.425**	.609**	.173**	1
	Sig. (2-tailed)	.000	.000	.000	.006	
	N	250	250	250	250	250

** . Correlation is significant at the 0.01 level (2-tailed).

* . Correlation is significant at the 0.05 level (2-tailed).

* /Correlation Matrix/

CORRELATIONS

```

/VARIABLES=age education_years experience_years training_hours annual_salary_k
/PRINT=TWOTAIL NOSIG LOWER
/MISSING=PAIRWISE.

```

Correlations

Pearson Correlation Matrix

	Age at time of extract (Years)	Total years of completed formal education (Years)	Total years of professional work experience (Years)	Company- funded training received in the last 12 months (Hours)	Gross annual salary (USD thousands)
Age at time of extract (Years)	--				
Total years of completed formal education (Years)	.420**	--			
Total years of professional work experience (Years)	.785**	.007	--		
Company-funded training received in the last 12 months (Hours)	.098	-.031	.126*	--	
Gross annual salary (USD thousands)	.624**	.425**	.609**	.173**	--

** Correlation is significant at the 0.01 level (2-tailed).

* Correlation is significant at the 0.05 level (2-tailed).

* I had to put an image of the correlation matrix here because SPSS kept breaking the format for and I couldn't export the whole table, which was utterly annoying.

```

OUTPUT MODIFY
  /SELECT TABLES
  /IF COMMANDS=["Correlations(LAST)"]
    SUBTYPES=["Correlations"]
  /TABLECELLS SELECT=[SIGNIFICANCE COUNT]
    SELECTDIMENSION=ROWS
    APPLYTO=ROW
    HIDE=YES
/TABLECELLS
  SELECT=[TITLE]
  REPLACE="Pearson Correlation Matrix"
/TABLECELLS
  SELECT=["CORRELATION"]
  SELECTDIMENSION=ROWS
  APPLYTO=ROWHEADER
  HIDE=YES.

* /Task 4/

REGRESSION
  /MISSING LISTWISE
  /STATISTICS COEFF OUTS CI(95) R ANOVA
  /CRITERIA=PIN(.05) POUT(.10)
  /NOORIGIN
  /DEPENDENT annual_salary_k
  /METHOD=ENTER education_years experience_years training_hours.

```

Regression

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	Company-funded training received in the last 12 months (Hours), Total years of completed formal education (Years), Total years of professional work experience (Years) ^b	.	Enter

a. Dependent Variable: Gross annual salary (USD thousands)

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.749 ^a	.561	.555	6.7495

a. Predictors: (Constant), Company-funded training received in the last 12 months (Hours), Total years of completed formal education (Years), Total years of professional work experience (Years)

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	14293.667	3	4764.556	104.588	.000 ^b
	Residual	11206.636	246	45.555		
	Total	25500.302	249			

a. Dependent Variable: Gross annual salary (USD thousands)

b. Predictors: (Constant), Company-funded training received in the last 12 months (Hours), Total years of completed formal education (Years), Total years of professional work experience (Years)

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B	
		B	Std. Error	Beta			Lower Bound	Upper Bound
1	(Constant)	19.541	3.266		5.983	.000	13.108	25.974
	Total years of completed formal education (Years)	1.895	.189	.425	10.038	.000	1.523	2.267
	Total years of professional work experience (Years)	1.217	.088	.592	13.899	.000	1.044	1.389
	Company-funded training received in the last 12 months (Hours)	.076	.029	.111	2.604	.010	.018	.133

a. Dependent Variable: Gross annual salary (USD thousands)

* Comment-1: These three factors (formal education, years of professional work experience, and hours of company-funded training) together account for 56.1% of the variation in salary, which we found from the value of R square.

* Comment-2: The estimated change in annual salary associated with one additional year of education, holding experience and training constant is 1895 USD on average.

* Comment-3: The estimated change in annual salary associated with one additional year of experience, holding year of education and training constant is 1217 USD on average.

* Comment-4: From regression analysis, we can see that the estimated change in annual salary associated with one additional hour of company-funded training, holding other factors constant is 76 USD on average.

* For this specific factor, $t(246) = 2.604$, $p = .010$, which is statistically significant at 5% level of significance.

* Thus we can conclude that, even though the increase in salary for one additional hour of company-funded training isn't that much compared to one unit of change in other factors, it's still statistically significant.

* Which means the money currently spent on training is associated with measurably higher pay.

```
SAVE OUTFILE='E:\Statistical
Softwares\SPSS\CUSPC-SPSS\Assignments\Assignment-2\Assignment-2_Dataset.sav'
/COMPRESSED.
```

Appendix:

* Encoding: UTF-8.

PRESERVE.
SET DECIMAL DOT.

* Since I messed up while generating the output of my first assignment, I'll be properly segmenting my code so that the final output is clean and reproducible using whatever syntax I used during data analysis.

* /Step-1: Data loading/

```
GET DATA /TYPE=TXT
  /FILE="E:\Statistical Softwares\SPSS\CUSPC-SPSS\Class-7\salary regression data.csv"
  /ENCODING='UTF8'
  /DELIMITERS=", "
  /QUALIFIER='"'
  /ARRANGEMENT=DELIMITED
  /FIRSTCASE=2
  /DATATYPEMIN PERCENTAGE=95.0
  /VARIABLES=
    employee_id AUTO
    age AUTO
    education_years AUTO
    experience_years AUTO
    training_hours AUTO
    region AUTO
    annual_salary_k AUTO
  /MAP.
RESTORE.
CACHE.
EXECUTE.
DATASET NAME DataSet1 WINDOW=FRONT.
```

* /Step-2: Data labelling/

```
VARIABLE LABELS
  employee_id 'Anonymised employee identifier'
  age 'Age at time of extract (Years)'
  education_years 'Total years of completed formal education (Years)'
  experience_years 'Total years of professional work experience (Years)'
  training_hours 'Company-funded training received in the last 12 months (Hours)'
  region 'Office location type (Urban/Suburban/Rural)'
  annual_salary_k 'Gross annual salary (USD thousands)'.
EXECUTE.

VARIABLE LEVEL employee_id (NOMINAL).
EXECUTE.
```

* /Step-3: Data Checking/

```
DATASET ACTIVATE DataSet1.
FREQUENCIES VARIABLES=employee_id age education_years experience_years training_hours
region
  annual_salary_k
  /FORMAT=NOTABLE
  /STATISTICS=RANGE MINIMUM MAXIMUM
  /ORDER=ANALYSIS.

FREQUENCIES VARIABLES=region
  /ORDER=ANALYSIS.
```

* All 250 observations are present in the given dataset, and no impossible values were found. We can safely proceed to the next steps of analysis.

* /Task 1/

.

```

* Custom Tables.
CTABLES
  /VLABELS VARIABLES=age education_years experience_years training_hours
  annual_salary_k
  DISPLAY=LABEL
  /TABLE age [COUNT 'Complete cases (n)' F40.0, MINIMUM, MAXIMUM, MEAN, MEDIAN, STDDEV]
+
  education_years [COUNT 'Complete cases (n)' F40.0, MINIMUM, MAXIMUM, MEAN, MEDIAN,
STDDEV] +
  experience_years [COUNT 'Complete cases (n)' F40.0, MINIMUM, MAXIMUM, MEAN, MEDIAN,
STDDEV] +
  training_hours [COUNT 'Complete cases (n)' F40.0, MINIMUM, MAXIMUM, MEAN, MEDIAN,
STDDEV] +
  annual_salary_k [COUNT 'Complete cases (n)' F40.0, MINIMUM, MAXIMUM, MEAN, MEDIAN,
STDDEV]
  /CRITERIA CILEVEL=95.

* /Task 2/

* /Graph-2a/

* Chart Builder.
GGRAPH
  /GRAPHDATASET NAME="graphdataset" VARIABLES=annual_salary_k MISSING=LISTWISE
REPORTMISSING=NO
  /GRAPHSPEC SOURCE=INLINE
  /FRAME OUTER=NO INNER=NO
  /GRIDLINES XAXIS=NO YAXIS=YES
  /STYLE GRADIENT=NO.
BEGIN GPL
  SOURCE: s=userSource(id("graphdataset"))
  DATA: annual_salary_k=col(source(s), name("annual_salary_k"))
  GUIDE: axis(dim(1), label("Gross annual salary (USD thousands)"))
  GUIDE: axis(dim(2), label("Frequency"))
  GUIDE: text.title(label("Histogram of Gross annual salary (USD thousands)"))
  ELEMENT: interval(position(summary.count(bin.rect(annual_salary_k))),
    shape.interior(shape.square))
END GPL.

* /Graph-2b/

* Chart Builder.
GGRAPH
  /GRAPHDATASET NAME="graphdataset" VARIABLES=region COUNT()[name="COUNT"]
MISSING=LISTWISE
  REPORTMISSING=NO
  /GRAPHSPEC SOURCE=INLINE
  /FRAME OUTER=NO INNER=NO
  /GRIDLINES XAXIS=NO YAXIS=YES
  /STYLE GRADIENT=NO.
BEGIN GPL
  SOURCE: s=userSource(id("graphdataset"))
  DATA: region=col(source(s), name("region"), unit.category())
  DATA: COUNT=col(source(s), name("COUNT"))
  GUIDE: axis(dim(1), label("Office location type (Urban/Suburban/Rural)"))
  GUIDE: axis(dim(2), label("Percent"))
  GUIDE: text.title(label("Bar chart of Office location type (Urban/Suburban/Rural)"))
  SCALE: linear(dim(2), include(0))
  ELEMENT: interval(position(summary.percent(region*COUNT,
base.all(acrossPanels()))),
    shape.interior(shape.square))
END GPL.

* /Graph-2c/

* Chart Builder.
GGRAPH
  /GRAPHDATASET NAME="graphdataset" VARIABLES=region annual_salary_k MISSING=LISTWISE

```



```

REPORTMISSING=NO
/GRAPHSPEC SOURCE=INLINE
/FRAME OUTER=NO INNER=NO
/GRIDLINES XAXIS=NO YAXIS=YES
/STYLE GRADIENT=NO.
BEGIN GPL
SOURCE: s=userSource(id("graphdataset"))
DATA: region=col(source(s), name("region"), unit.category())
DATA: annual_salary_k=col(source(s), name("annual_salary_k"))
DATA: id=col(source(s), name("$CASENUM"), unit.category())
COORD: transpose()
GUIDE: axis(dim(1), label("Office location type (Urban/Suburban/Rural)"))
GUIDE: axis(dim(2), label("Gross annual salary (USD thousands)"))
GUIDE: text.title(label("Boxplot of Gross annual salary (USD thousands) split by
Office location type (Urban/Suburban/Rural)"))
SCALE: linear(dim(2), include(0))
ELEMENT: schema(position(bin.quantile.letter(region*annual_salary_k)), label(id))
END GPL

* /Graph-2d_1/

* Chart Builder.
GGRAPH
/GRAPHDATASET NAME="graphdataset" VARIABLES=education_years annual_salary_k
MISSING=LISTWISE
REPORTMISSING=NO
/GRAPHSPEC SOURCE=INLINE
/FITLINE TOTAL=YES SUBGROUP=NO
/FRAME OUTER=NO INNER=NO
/GRIDLINES XAXIS=NO YAXIS=YES
/STYLE GRADIENT=NO.
BEGIN GPL
SOURCE: s=userSource(id("graphdataset"))
DATA: education_years=col(source(s), name("education_years"))
DATA: annual_salary_k=col(source(s), name("annual_salary_k"))
GUIDE: axis(dim(1), label("Total years of completed formal education (Years)"))
GUIDE: axis(dim(2), label("Gross annual salary (USD thousands)"))
GUIDE: text.title(label("Scatter Plot of Gross annual salary (USD thousands) by Total
years of ",
"completed formal education (Years)"))
ELEMENT: point(position(education_years*annual_salary_k))
END GPL

* /Graph-2d_2/

* Chart Builder.
GGRAPH
/GRAPHDATASET NAME="graphdataset" VARIABLES=experience_years annual_salary_k
MISSING=LISTWISE
REPORTMISSING=NO
/GRAPHSPEC SOURCE=INLINE
/FITLINE TOTAL=YES SUBGROUP=NO
/FRAME OUTER=NO INNER=NO
/GRIDLINES XAXIS=NO YAXIS=YES
/STYLE GRADIENT=NO.
BEGIN GPL
SOURCE: s=userSource(id("graphdataset"))
DATA: experience_years=col(source(s), name("experience_years"))
DATA: annual_salary_k=col(source(s), name("annual_salary_k"))
GUIDE: axis(dim(1), label("Total years of professional work experience (Years)"))
GUIDE: axis(dim(2), label("Gross annual salary (USD thousands)"))
GUIDE: text.title(label("Scatter Plot of Gross annual salary (USD thousands) by Total
years of ",
"professional work experience (Years)"))
ELEMENT: point(position(experience_years*annual_salary_k))
END GPL

* /Graph-2d_3/

```

```

* Chart Builder.
GGRAPH
  /GRAPHDATASET NAME="graphdataset" VARIABLES=training_hours annual_salary_k
MISSING=LISTWISE
  REPORTMISSING=NO
  /GRAPHSPEC SOURCE=INLINE
  /FITLINE TOTAL=YES SUBGROUP=NO
  /FRAME OUTER=NO INNER=NO
  /GRIDLINES XAXIS=NO YAXIS=YES
  /STYLE GRADIENT=NO.
BEGIN GPL
  SOURCE: s=userSource(id("graphdataset"))
  DATA: training_hours=col(source(s), name("training_hours"))
  DATA: annual_salary_k=col(source(s), name("annual_salary_k"))
  GUIDE: axis(dim(1), label("Company-funded training received in the last 12 months
(Hours)"))
  GUIDE: axis(dim(2), label("Gross annual salary (USD thousands)"))
  GUIDE: text.title(label("Scatter Plot of Gross annual salary (USD thousands) by
Company-funded ",
  "training received in the last 12 months (Hours)"))
  ELEMENT: point(position(training_hours*annual_salary_k))
END GPL.

```

* /Task 3/

* /Main Table/

```

CORRELATIONS
  /VARIABLES=age education_years experience_years training_hours annual_salary_k
  /PRINT=TWOTAIL NOSIG FULL
  /MISSING=PAIRWISE.

```

* /Correlation Matrix/

```

CORRELATIONS
  /VARIABLES=age education_years experience_years training_hours annual_salary_k
  /PRINT=TWOTAIL NOSIG LOWER
  /MISSING=PAIRWISE.

```

```

OUTPUT MODIFY
  /SELECT TABLES
  /IF COMMANDS=["Correlations(LAST)"]
    SUBTYPES=["Correlations"]
  /TABLECELLS SELECT=[SIGNIFICANCE COUNT]
    SELECTDIMENSION=ROWS
    APPLYTO=ROW
    HIDE=YES
/TABLECELLS
  SELECT=[TITLE]
  REPLACE="Pearson Correlation Matrix"
/TABLECELLS
  SELECT=["CORRELATION"]
  SELECTDIMENSION=ROWS
  APPLYTO=ROWHEADER
  HIDE=YES.

```

* /Task 4/

```

REGRESSION
  /MISSING LISTWISE
  /STATISTICS COEFF OUTS CI(95) R ANOVA
  /CRITERIA=PIN(.05) POUT(.10)
  /NOORIGIN
  /DEPENDENT annual_salary_k
  /METHOD=ENTER education_years experience_years training_hours.

```

* Comment-1: These three factors (formal education, years of professional work

experience, and hours of company-funded training) together account for 56.1% of the variation in salary, which we found from the value of R square.

* Comment-2: The estimated change in annual salary associated with one additional year of education, holding experience and training constant is 1895 USD on average.

* Comment-3: The estimated change in annual salary associated with one additional year of experience, holding year of education and training constant is 1217 USD on average.

* Comment-4: From regression analysis, we can see that the estimated change in annual salary associated with one additional hour of company-funded training, holding other factors constant is 76 USD on average.

* For this specific factor, $t(246) = 2.604$, $p = .010$, which is statistically significant at 5% level of significance.

* Thus we can conclude that, even though the increase in salary for one additional hour of company-funded training isn't that much compared to one unit of change in other factors, it's still statistically significant.

* Which means the money currently spent on training is associated with measurably higher pay.

```
SAVE OUTFILE='E:\Statistical
Softwares\SPSS\CUSPC-SPSS\Assignments\Assignment-2\Assignment-2_Dataset.sav'
/COMPRESSED.
```